

OS Unit 4 — Memory Management & Virtual Memory Complete Notes

Operating Systems (OS) · Unit 4 · Memory Management & Virtual Memory · 2025

Overview

These notes cover **Memory Management & Virtual Memory**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

1. Contiguous Allocation

First fit, best fit and worst fit allocate memory partitions, causing external fragmentation.

Why it matters: Contiguous Allocation appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of contiguous allocation in one line.
- A small example showing contiguous allocation in practice.
- Common mistakes students make and how to avoid them.

2. Paging

Paging divides memory into fixed-size frames and uses page tables to translate addresses.

Why it matters: Paging appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of paging in one line.
- A small example showing paging in practice.
- Common mistakes students make and how to avoid them.

3. Translation Lookaside Buffer

The TLB caches page table entries to make address translation fast.

Why it matters: Translation Lookaside Buffer appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of translation lookaside buffer in one line.
- A small example showing translation lookaside buffer in practice.
- Common mistakes students make and how to avoid them.

4. Segmentation

Segmentation divides a program into logical segments such as code, stack and heap.

Why it matters: Segmentation appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of segmentation in one line.
- A small example showing segmentation in practice.
- Common mistakes students make and how to avoid them.

5. Demand Paging

Pages are loaded only when referenced, triggering page faults otherwise.

Why it matters: Demand Paging appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of demand paging in one line.
- A small example showing demand paging in practice.
- Common mistakes students make and how to avoid them.

6. Page Replacement Algorithms

FIFO, optimal and LRU choose victim pages; FIFO can exhibit Belady's anomaly.

Why it matters: Page Replacement Algorithms appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of page replacement algorithms in one line.
- A small example showing page replacement algorithms in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Contiguous Allocation · Paging · Translation Lookaside Buffer · Segmentation · Demand Paging · Page Replacement Algorithms